

# Ontological Hierarchy of Energy Reduction The Levels of Fundamental Reality in Natural Units From Time-Mass Duality to Universal Energy Field

## Contents

## Abstract

This work examines the ontological hierarchy of T0 theory under the paradigm of natural units, where through time-mass duality  $T \cdot m = 1$  all physical quantities can be reduced to energy. The central insight: There exist **five ontological levels of reduction**, ranging from the most fundamental (universal energy field) to observable physics. Each level emerges from the underlying one through mathematical necessity. The analysis shows: (1) **Level 0 – Absolute Foundation**: The universal energy field  $E_{\text{Field}}(x, t)$  with wave equation  $\square E = 0$ . (2) **Level 1 – Time-Mass Duality**:  $T(x, t) \cdot m(x, t) = 1$  in natural units. (3) **Level 2 – Geometric Parameters**:  $\xi = 4/30000$  and 4D torsion structure. (4) **Level 3 – Effective Field Theory**: Modified laws with  $\sim 1\text{--}2\%$  corrections. (5) **Level 4 – SI Units Physics**: Classical observation level with  $c, \hbar, G$  as separate constants. Narrative integration occurs through upward propagation: From the fundamental energy field emerges duality, from that geometry, from that effective laws, from that classical physics.

## .1 Introduction: The Reduction Program

### The Central Question

#### Fundamental Question

If in natural units ( $\hbar = c = 1$ ) through time-mass duality everything can be reduced to energy, which ontological levels exist, and how do they organize themselves hierarchically?

Put differently: What are the **depths of reality** when we systematically descend from human conventions (SI units) to fundamental structures (energy field)?

### The Dimensional Reduction

In natural units:

$$\hbar = c = 1 \quad \Rightarrow \quad [L] = [T] = [E^{-1}], \quad [M] = [E] \quad (1)$$

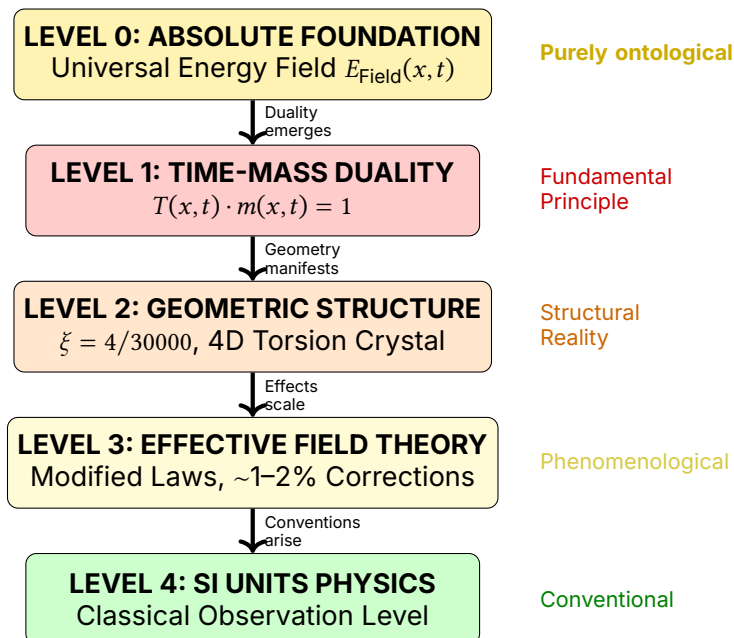
**Consequence:** All physical quantities are reduced to **one dimension** – energy!

| Quantity    | SI Units | Natural Units |
|-------------|----------|---------------|
| Length      | m        | $E^{-1}$      |
| Time        | s        | $E^{-1}$      |
| Mass        | kg       | $E$           |
| Temperature | K        | $E$           |
| Charge      | C        | dimensionless |
| Energy      | J        | $E$           |

**Table 1:** Dimensional reduction in natural units

## .2 The Five Ontological Levels

### Hierarchy Overview



## .3 Level 0: The Absolute Foundation

### Ontological Description

### The Most Fundamental Reality

At the deepest level exists:

A Universal Energy Field  $E_{\text{Field}}(x, t)$

This field is:

- **Non-dual:** No separation into space/time/mass
- **Self-evident:** Requires no further concepts
- **Dynamic:** Obeys the wave equation
- **Universal:** Permeates the entire universe

### The Fundamental Equation

$$\square E_{\text{Field}}(x, t) = 0$$

(2)

where  $\square = \frac{\partial^2}{\partial t^2} - \nabla^2$  is the d'Alembert operator.

**Physical meaning:**

- Energy propagates as wave
- No sources or sinks at fundamental level
- Completely deterministic
- Local in space and time

### Why is this fundamental?

#### Justification of Fundamentality

The energy field is fundamental because:

##### 1. Minimal assumptions:

- Only one field
- Only one equation
- No free parameters (in natural units)

##### 2. Maximal explanatory power:

- All other concepts emerge from it
- Space = configuration space of the field
- Time = evolution of the field
- Mass = field excitation

##### 3. Mathematical elegance:

- Linear (superposition valid)
- Lorentz invariant
- Energy conserving

### Ontological Status

What exists:

- The energy field  $E_{\text{Field}}(x, t)$
- Its configuration at each time
- Its evolution dynamics
- **What doesn't exist** (at this level):
- Separate time as independent entity
- Separate mass as substance
- Particles as fundamental objects
- Space as empty container

## .4 Level 1: Time-Mass Duality

### Emergence of Duality

From the fundamental energy field emerges the first structuring:

#### Time-Mass Duality

In natural units holds the fundamental relationship:

$$T(x, t) \cdot m(x, t) = 1 \quad (3)$$

This is equivalent to:

$$T(x, t) = \frac{1}{m(x, t)} = \frac{1}{E(x, t)} \quad (4)$$

### Mathematical Derivation

From the Heisenberg uncertainty principle:

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2} \quad (5)$$

In natural units ( $\hbar = 1$ ):

$$\Delta E \cdot \Delta t \geq \frac{1}{2} \quad (6)$$

In the limit  $\Delta \rightarrow 0$ :

$$E \cdot T = 1 \quad \Leftrightarrow \quad m \cdot T = 1 \quad (7)$$

### The Intrinsic Time Field

The duality manifests as a field:

$$T(x, t) = \frac{1}{\max(m(x, t), \omega)} \quad (8)$$

**Dimensional verification:**

$$[T(x, t)] = [E^{-1}] \quad (9)$$

$$[m(x, t)] = [E] \quad (10)$$

$$[T \cdot m] = [E^{-1}] \cdot [E] = [1] \quad (11)$$

## Ontological Status

At this level exist:

- Time as **field quantity**  $T(x, t)$  (not as parameter)
- Mass as **field quantity**  $m(x, t)$  (not as substance)
- Their reciprocal relationship as **fundamental law**

**Physical meaning:**

- Time varies with energy:  $T \propto 1/E$
- Mass varies with energy:  $m \propto E$
- Both are **aspects of the energy field**

## Reduction to Energy

In natural units:

$$E = m \quad (\text{Energy} = \text{Mass}) \quad (12)$$

$$E = \omega \quad (\text{Energy} = \text{Frequency}) \quad (13)$$

$$E = 1/T \quad (\text{Energy} = \text{inverse time}) \quad (14)$$

$$E = 1/L \quad (\text{Energy} = \text{inverse length}) \quad (15)$$

**Everything is energy in various manifestations!**

## .5 Level 2: Geometric Structure

### Emergence of Geometry

From time-mass duality emerges geometric structure:

#### Geometric Manifestation

The duality manifests geometrically as:

- **Parameter:**  $\xi = \frac{4}{30000} = 1.333 \times 10^{-4}$
- **Structure:** 4D torsion crystal
- **Scale:** Sub-Planck granulation  $\Lambda_0 = \xi \cdot \ell_P$
- **Symmetry:** Pentagonal breaking via golden ratio  $\varphi$

### The Field Equation

The time-mass field obeys:

$$\nabla^2 m(x, t) = 4\pi G \rho(x, t) \cdot m(x, t) \quad (16)$$

**Dimensional verification** (natural units):

$$[\nabla^2 m] = [E^2] \cdot [E] = [E^3] \quad (17)$$

$$[4\pi G \rho m] = [1] \cdot [E^{-2}] \cdot [E^4] \cdot [E] = [E^3] \quad (18)$$

## Geometric Parameters

From the field equation follow:

$$\beta = \frac{2Gm}{r} = \frac{2m}{r} \quad (\text{in nat. units with } G = 1) \quad (19)$$

$$\xi_{\text{geom}} = 2\sqrt{G} \cdot m = 2m \quad (\text{geometric parameter}) \quad (20)$$

## The 4D Torsion Structure

**Topology:**

$$\mathcal{M}_{\text{fund}} = \mathbb{R}^3 \times S_{\text{Comp}}^1 \quad (21)$$

where:

- $\mathbb{R}^3$  = observable 3D space
- $S_{\text{Comp}}^1$  = compactified 4th dimension with radius  $r_4 = \xi \cdot \ell_P$

## Ontological Status

**At this level exist:**

- Geometric structure as **emergent property** of duality
- Parameter  $\xi$  as **manifestation** of 4D structure
- Torsion as **twisting** of compact dimension
- **Not yet existent** (only higher levels):
- Separate constants  $c, \hbar, G$
- Particles as distinct objects
- Classical trajectories

## .6 Level 3: Effective Field Theory

### Emergence of Phenomenological Laws

From geometric structure emerge measurable effects:

#### Effective Description

At measurable scales ( $\ell \gg \Lambda_0$ ) we see:

- Modified force laws with  $\xi$ -corrections
- Fractal dimension  $D_f = 3 - \xi$
- Anomalous moments with  $\sim 2\%$  deviation
- Geometric constant predictions

## Modified Laws

**Coulomb's law:**

$$F_{\text{Coulomb}} \propto \frac{1}{r^{1+\xi}} \approx \frac{1}{r^2} \left( 1 - \xi \ln \frac{r}{\ell_P} \right) \quad (22)$$

**Gravitational potential:**

$$\Phi(r) = -\frac{Gm}{r}(1 + \kappa r) \quad (23)$$

**Fine structure constant:**

$$\alpha^{-1} = \pi^4 \cdot \sqrt{2} \approx 137.76 \quad (24)$$

## Correction Factors

Over many orders of magnitude,  $\xi$  accumulates:

$$K_{\text{frak}} = 1 - 100\xi \approx 0.9867 \quad (25)$$

This leads to  $\sim 1.33\%$  corrections in observables.

## Ontological Status

**At this level exist:**

- Effective laws as **approximations** of geometry
- Measurable deviations from Standard Model
- Phenomenological parameters (not yet  $c, \hbar, G$  separate)

**Characteristics:**

- **Not fundamental**, but practically relevant
- **Emergent** from deeper levels
- **Approximative** with defined accuracy

## .7 Level 4: SI Units Physics

### Emergence of Conventions

From effective theory emerge human conventions:

#### Conventional Physics

For practical purposes we introduce:

- Separate constants:  $c = 299\,792\,458$  m/s,  $\hbar = 1.055 \times 10^{-34}$  Js
- Separate units: Meter, kilogram, second
- Separate quantities: Energy  $\neq$  mass  $\neq$  time

**This is the level of human measurements!**

### Back Translation

From natural to SI units:

$$E \text{ (nat.)} \rightarrow E \text{ (SI)} = E \cdot (\hbar c) \quad (26)$$

$$m \text{ (nat.)} \rightarrow m \text{ (SI)} = m \cdot \frac{\hbar}{c^2} \quad (27)$$

$$T \text{ (nat.)} \rightarrow T \text{ (SI)} = T \cdot \frac{\hbar}{c^2} \quad (28)$$



## Ontological Status

At this level exist:

- Human conventions as **measurement tools**
- Separate concepts for practical applications
- Classical approximations for everyday physics

**Characteristics:**

- **Not fundamental**, but conventional
- **Useful** for technology and experiments
- **Obscures** the deeper unity of physics

## .8 Hierarchy Summary

### The Complete Chain

| Level | Description       | What exists              | Status                 |
|-------|-------------------|--------------------------|------------------------|
| 0     | Energy Field      | $E_{\text{Field}}(x, t)$ | Absolutely fundamental |
| 1     | Time-Mass Duality | $T \cdot m = 1$          | First emergence        |
| 2     | Geometry          | $\xi$ , 4D Torsion       | Structural reality     |
| 3     | Effective Theory  | Modified Laws            | Phenomenological       |
| 4     | SI Physics        | $c, \hbar, G$ separate   | Conventional           |

**Table 2:** The five ontological levels

### Causal Relationships

$$\text{Level 0} \Rightarrow \text{Level 1} \Rightarrow \text{Level 2} \Rightarrow \text{Level 3} \Rightarrow \text{Level 4} \quad (29)$$

where  $\Rightarrow$  means: "determines" or "allows to emerge"

### Reduction to Energy

**At all levels holds in natural units:**

$$[X] = [E]^n$$

for some  $n \in \mathbb{Z}$

**Everything is energy!**

## .9 Narrative Integration

### Bottom-Up: The Emergence Narrative

#### The Story of Reality

**LEVEL 0 – In the beginning was the field:**

There exists a universal energy field  $E_{\text{Field}}(x, t)$  that obeys the wave equation  $\square E = 0$ .  
Nothing else exists – only this one field.

⇓

#### **LEVEL 1 – Duality emerges:**

From the quantum nature of the field ( $\Delta E \cdot \Delta t \geq \hbar/2$ ) emerges time-mass duality:  $T \cdot m = 1$ . Time is no longer parameter, but field!

⇓

#### **LEVEL 2 – Geometry manifests:**

The duality manifests geometrically: 4D torsion crystal with parameter  $\xi = 4/30000$ , compact 4th dimension at sub-Planck scale.

⇓

#### **LEVEL 3 – Effects scale:**

At measurable scales we see modified laws: Coulomb  $\propto 1/r^{1+\xi}$ , anomalous moments with  $\sim 2\%$  deviation, geometric constants.

⇓

#### **LEVEL 4 – Conventions arise:**

Humans introduce SI units: meter, kilogram, second. They artificially separate  $c, \hbar, G$ . The deeper unity is obscured.

## **Top-Down: The Reduction Narrative**

### **The Path to Fundamentality**

#### **START: SI Physics (Level 4)**

We begin with separate concepts: energy, mass, time, length. We have many constants:  $c, \hbar, G, k_B, \dots$

⇓ *Simplification*

#### **Natural Units (Level 3)**

We set  $c = \hbar = 1$ . Suddenly: energy = mass, time = inverse energy. Everything becomes simpler!

⇓ *Deeper analysis*

#### **Geometric Structure (Level 2)**

We recognize: The simplicity comes from 4D geometry. Parameter  $\xi$  encodes everything. Torsion explains mass!

⇓ *Ultimate reduction*

#### **Time-Mass Duality (Level 1)**

We understand: Time and mass are dual,  $T \cdot m = 1$ . Both are aspects of energy!

⇓ *Fundamental truth*

#### **Universal Energy Field (Level 0)**

At the foundation: One field, one equation. Everything else emerges.

## .10 Comparison of Both Descriptions

### 4D Torsion Crystal vs. Energy Reduction

| 4D Torsion Crystal (Level 2)                                             | Energy Reduction (Level 0–1)                                                       |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Geometric perspective<br>Intuitive: Twisting<br>4 dimensions topological | Field-theoretic perspective<br>Abstract: Duality<br>1 dimension (energy) reductive |
| Torsion as cause<br>Sub-Planck structure primary                         | Field excitation as cause<br>Wave equation primary                                 |
| <b>BOTH describe the same reality!</b>                                   |                                                                                    |
| Level 2 in hierarchy<br>Emerges from Level 1<br>Geometrically manifest   | Level 0–1 in hierarchy<br>Fundamental for Level 2<br>Energetically fundamental     |

**Table 3:** Complementary descriptions

### Ontological Classification

#### How do both fit in?

##### Energy Reduction (Level 0–1):

- **More fundamental** – goes deeper
- **More abstract** – less intuitive
- **More universal** – holds without restrictions

##### 4D Torsion Crystal (Level 2):

- **Emergent** – follows from Level 1
- **More intuitive** – geometrically visualizable
- **Structural** – manifests duality

##### Relationship:

Energy Field (Level 0)  $\xrightarrow{\text{creates}}$  Duality (Level 1)  $\xrightarrow{\text{manifests}}$  Geometry (Level 2)

### Why Both Descriptions Coexist

#### Complementarity

Analogous to wave-particle duality in quantum mechanics:

##### Energy Reduction:

- Like wave description
- Fundamental, but abstract
- Mathematically elegant
- Hard to visualize

##### 4D Geometry:

- Like particle description
- Emergent, but intuitive
- Geometrically intuitive
- Practically useful

**Both are valid**, describing different aspects of the same reality!

## .11 Practical Consequences

### For Calculations

#### Which level to choose?

##### Level 0–1 (Energy Reduction):

- Theoretical derivations
- Fundamental principles
- Symmetry arguments
- Conceptual clarity

##### Level 2 (Geometry):

- Visual explanations
- Particle masses
- Structural predictions
- Narrative presentations

##### Level 3 (Effective):

- Experimental predictions
- Comparison with data
- Phenomenology

##### Level 4 (SI):

- Practical measurements
- Technology
- Everyday applications

### For Communication

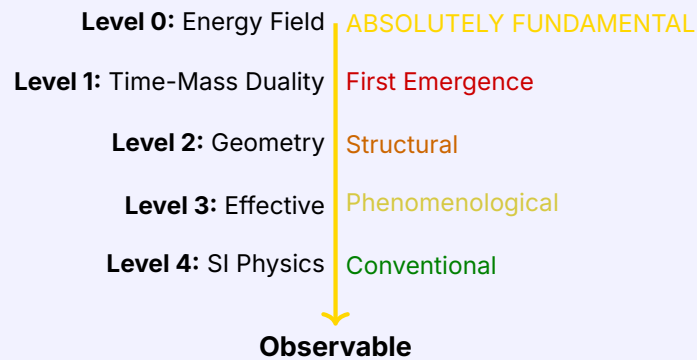
| Target Audience | Preferred Level     | Reason      |
|-----------------|---------------------|-------------|
| Laypeople       | Level 4 (SI)        | Familiar    |
| Students        | Level 3 (Effective) | Learnable   |
| Physicists      | Level 2 (Geometry)  | Intuitive   |
| Theorists       | Level 1 (Duality)   | Fundamental |
| Philosophers    | Level 0 (Field)     | Ontological |

**Table 4:** Level choice by target audience

## .12 Conclusion

### Main Result

T0 theory possesses a clear **\*\*five-level ontological hierarchy\*\***:



Through natural units, everything is reduced to energy.  
 The 4D geometry is Level 2 – emergent from duality (Level 1).  
 The universal energy field (Level 0) is the absolute foundation.

### The Ultimate Reduction

#### The Truth of Physics

##### Everything is Energy

Space, time, mass, charge, forces, particles – all these are only different **\*\*manifestations of a single universal energy field\*\***.

In natural units this becomes mathematically explicit:

$$[X] = [E]^n \quad \text{for every physical quantity } X \quad (30)$$

The time-mass duality  $T \cdot m = 1$  is the key to this insight.

The 4D torsion crystal is the geometric manifestation of this fundamental truth.